

PRELIMINARY DATASET REPORT

Dataset Identification

The given archive is the public dataset of the LG 18650HG2 lithium-ion battery tests, recorded at McMaster University under the supervision of Dr. Phillip Kollmeyer. One physical 3 Ah cell, and not an array of cells, was charged and discharged in a battery cyclor within a thermal chamber across six ambient temperature set points in sequence: -20°C, -10°C, 0°C, 10°C, 25°C, and 40°C. The battery was subjected to one of four sets of protocols at each temperature set point: HPPC (hybrid pulse power characterization) pulses, discharging at C/20 and higher rates, capacity-check test (Cap_1C), and four different drive cycles – UDDS, HWFET, LA92, US06, or random combinations of those drives with charge intervals between them.

It is a critical detail that needs to be considered in regard to all further analysis because it is a public and thoroughly characterized benchmark, not proprietary telemetry from an active device, making it great for creating and testing the modeling pipeline but requiring the understanding of its protocol specifics (given below).

Raw Data Preservation

The archive was not opened up for writing. All the figures presented in this report have been obtained by directly reading the unaltered document, and no rows, columns, or files have been stripped out, resampled, or overwritten during this phase of processing. This is important because the issues detected further down the line are only credible as "true" data quality issues if we can assure ourselves that nothing was altered prior to running the audit.

Files and row Inventory

The 208 CSV files are comprised of n20degC (35), n10degC (38), 0degC (32), 10degC (32), 25degC (36), and 40degC (35) folders respectively. These numbers are not evenly distributed which is expected given that different temperatures received different numbers of Charge and Mixed sub-segment files (being session specific, not quantity specific).

The total number of rows summed up to 4,954,917 which is an extremely large set of time-series data. However, that does not mean the data is evenly distributed either. Given that drive-cycle files are typically logged at ~0.1s resolution, a single UDDS or LA92 drive can easily contain more than 100,000 rows of information. Whereas long discharge (C/20) or charge (tapering) tests will have a much lower row per real-time second ratio.

I would not recommend any sampling procedure based on simply randomly selecting N rows from the entire data set because such a procedure would overwhelm the sample with drive-cycle data.

There are two types of CSV files in terms of column specification:

- 1) 207 files with a 15-column layout which includes a phantom (empty) column at the end due to an extra comma at the end of the header row in the original cyclers export; and
- 2) one file (549_Charge.csv) that has one fewer column due to not having that extra comma at the end of the header row. In reality, that file is a regular charge test: metadata block, measurement columns, and units row are identical to the rest of the files. The only difference is that the last (empty) column is omitted.

As a result, simply dropping the last column will make this file identical to the rest in terms of column specification (resulting in 14 columns): Time Stamp, Step, Status, Prog Time, Step Time, Cycle, Cycle Level, Procedure, Voltage, Current, Temperature, Capacity, WhAccu, Cnt.

Data Quality

Duplicates: 0 exact copy files detected, after comparing the full byte content of every one of the 208 files. Not a single file was logged twice or accidentally saved in the wrong folder.

Missing values: 19 nulls in the Voltage, Current, Temperature, Capacity, WhAccu, and Cnt columns, 3 or 4 in each (0.0004% of 4.95 million rows). Negligible in aggregate, but it is still critical to provide a precise figure – the next step in the analysis will require an imputation procedure not for all-null columns but for specific rows identified during the cleaning process.

Voltage range outlier – the most notable discovery, one row in each file contains Voltage value of 0.000V right in the middle of a discharge cycle:

- 590_Charge16.csv (0degC folder): min 0.000V, max 4.200V
- 552_Charge16.csv (25degC folder): min 0.000V, max 4.200V
- 557_Charge10.csv (40degC folder): min 0.000V, max 3.393V

A real lithium-ion battery is not supposed to decline in voltage – not to hit 0V in the middle of a discharge cycle and then climb back up to 4.2V/3.4V, depending on the group. Presumably, a 0V reading is linked to a connector dropout – either during the logging procedure itself or one of the measurements was simply not recorded at a particular timestamp. Either way, the three anomalous rows should be removed during the cleaning process – no imputation is required, since the data has a clear issue that needs to be manually resolved.

Test type coverage

All 12 test-type families (Charge, Mixed, Cap_1C, HPPC, C20DisCh, Dis_0p5C, Dis_2C, UDDS, HWFET, LA, US, and PausCycl) are present in every one of the six temperature folders, with no category missing in any. This is indeed a clean result – regardless of what approach to modeling is used, there is no temperature regime where a whole class of driving/testing behavior would be missing.

Yet, the crosstab also shows that Charge and Mixed files are vastly outnumbered by the others (14–19 Charge files and 7–8 Mixed files per folder compared to exactly 1 file each for HPPC, Cap_1C, C20DisCh, UDDS, HWFET, LA92, and US06). This is not really a case of “12 independent test designs of equal weight” – it is a few single-run reference tests plus numerous numbered sub-segments of the charge/recharge and randomized-mix protocols. Any sampling or evaluation split that takes place per file needs to take that into account by weighting appropriately according to what a file actually represents.

Critical Structure finding

25degC	:	2018-10-25 04:08:59	→	2018-10-31 19:43:45
40degC	:	2018-11-02 15:00:50	→	2018-11-08 20:14:22
10degC	:	2018-11-12 17:21:50	→	2018-11-26 12:40:49
0degC	:	2018-11-26 19:22:05	→	2018-12-02 19:38:12
n10degC	:	2018-12-04 19:46:56	→	2018-12-19 00:48:05
n20degC	:	2018-12-21 11:10:46	→	2018-12-27 01:02:51

The six "temperature" conditions were tested on the same cell, sequentially, over the course of about 64 days; that is, the temperature was ramped down from 25C to 40C to 10C to 0C to -10C to -20C, not on six different cells and not in a randomized order. The effects of temperature are therefore completely confounded with the effects of aging (or more accurately, with the passage of time) -- any observed difference in voltage or capacity between the different temperature "conditions" could be due to temperature, due to aging (since the cell was aged for a substantial period of time, especially at 25C and 40C), or some combination of the two, and it is impossible to tell which based solely on the given data. This needs to be stated clearly as a limitation of the model, and it has implications for how any train/test split should be done (a naive approach, which does not take temperature into account, would result in the test set containing much older cells than the training set).

Charging Session Run near room Temperature, regardless of folder label

Grouping the mean case temperature by whether a particular file is a Charge file or not reveals an interesting pattern: non-charge (drive-cycle/discharge) files track their folder's nominal setpoint relatively closely -15.8 C in n20degC, 31.0 C in 40degC – but Charge files are much narrower band (9-26 C), encompassing even the coldest and hottest folders. In other words, whatever the folder name (0degC, -20degC etc.), the description seems to apply to the ambient temperature during driving/discharge testing, while charging itself appears to take place at or around room temperature regardless of the folder the file is in. Any feature which assumes that "temperature folder = temperature when charging" would be wrong for every Charge file in the set.

- ❖ A preliminary examination of one set of same-cell charge files (611_Charge9.csv through 611_Charge17.csv, all n20degC) reveals that the starting voltages and amounts of charge passed during each session varied considerably – from nearly 0 Ah (a small topping charge between HPPC pulses beginning near 3.6-4.19 V) to almost 1.55 Ah for another session. Additionally, there is no obvious pattern to when each file was taken – some charge sessions appear to begin at nearly 0% state of charge and end at nearly 100%, but others followed different patterns. The “remaining charging time” labels – presumably the ultimate destination for this data – must be extracted from the data itself for each file individually, as it appears that no single file makes this information obvious at a glance.
- ❖ The schema indicates that the SoC or SoC% is not presented in any of the raw files, but Voltage, Current, Temperature, Ah, WhAccu, and Cnt are given instead. The SoC percentage has to be calculated using Coulomb counting principles by integrating the current, and dividing it by the capacity at full charge. It is essential to use the measured capacity (the Cap_1C in the file) for each cell, rather than the 3 Ah nominal capacity stated in the problem statement. This is because the actual capacity is lower than the rated one due to the aging of the battery, which was emphasized in section 6 of the paper.